

Answer ALL questions. ALL working should be shown.

1. Given a rotation matrix in the equivalent angle-axis form:

$${}^A_B R = \begin{bmatrix} k_x^2 v \theta + c \theta & k_x k_y v \theta - k_z s \theta & k_x k_z v \theta + k_y s \theta \\ k_x k_y v \theta + k_z s \theta & k_y^2 v \theta + c \theta & k_y k_z v \theta - k_x s \theta \\ k_x k_z v \theta - k_y s \theta & k_y k_z v \theta + k_x s \theta & k_z^2 v \theta + c \theta \end{bmatrix}$$

(a) Show that the (1,1) element of the matrix can be written in the quaternion form:

$$1 - 2\varepsilon_2^2 - 2\varepsilon_3^2$$

[5 marks]

From definition of quaternions:

$$\varepsilon_1 = k_x s \frac{\theta}{2}$$

$$\varepsilon_2 = k_y s \frac{\theta}{2}$$

$$\varepsilon_3 = k_z s \frac{\theta}{2}$$

$$\varepsilon_4 = c \frac{\theta}{2}$$

Where $\varepsilon_1^2 + \varepsilon_2^2 + \varepsilon_3^2 + \varepsilon_4^2 = 1$.

Using $\varepsilon_1 = k_x s \frac{\theta}{2}$, the (1,1) element of the matrix can be written as:

$$\begin{aligned} k_x^2 v \theta + c \theta &= \frac{\varepsilon_1^2}{s^2 \frac{\theta}{2}} v \theta + c \theta \\ &= \frac{\varepsilon_1^2}{s^2 \frac{\theta}{2}} (1 - c \theta) + c \theta \\ &= \frac{\varepsilon_1^2}{1 - c^2 \frac{\theta}{2}} (1 - c \theta) + c \theta \end{aligned}$$

Using trigonometric identity, $c \theta = 2c^2 \frac{\theta}{2} - 1$, we have:

$$\begin{aligned} k_x^2 v \theta + c \theta &= \frac{\varepsilon_1^2}{1 - c^2 \frac{\theta}{2}} \left(1 - 2c^2 \frac{\theta}{2} + 1 \right) + 2c^2 \frac{\theta}{2} - 1 \\ &= \frac{2\varepsilon_1^2}{1 - c^2 \frac{\theta}{2}} \left(1 - c^2 \frac{\theta}{2} \right) + 2c^2 \frac{\theta}{2} - 1 \\ &= 2\varepsilon_1^2 + 2c^2 \frac{\theta}{2} - 1 \end{aligned}$$

Using $\varepsilon_4 = c \frac{\theta}{2}$, we then have:

$$\begin{aligned} k_x^2 v \theta + c \theta &= 2\varepsilon_1^2 + 2\varepsilon_4^2 - 1 \\ &= 2(\varepsilon_1^2 + \varepsilon_4^2) - 1 \end{aligned}$$

But we know $\varepsilon_1^2 + \varepsilon_2^2 + \varepsilon_3^2 + \varepsilon_4^2 = 1$, thus:

$$\begin{aligned} k_x^2 v\theta + c\theta &= 2(1 - \varepsilon_2^2 - \varepsilon_3^2) - 1 \\ &= 1 - 2\varepsilon_2^2 - 2\varepsilon_3^2 \end{aligned}$$

Proven!

[5 marks for calculation]

(b) Show that the (2,1) element of the matrix can be written in the quaternion form:

$$2(\varepsilon_1 \varepsilon_2 + \varepsilon_3 \varepsilon_4)$$

[3 marks]

The (2,1) element of the matrix is $k_x k_y v\theta + k_z s\theta$. Again, by replacing the k terms using quaternions, we have:

$$\begin{aligned} k_x k_y v\theta + k_z s\theta &= \frac{\varepsilon_1}{s \frac{\theta}{2}} \cdot \frac{\varepsilon_2}{s \frac{\theta}{2}} \cdot (1 - c\theta) + \frac{\varepsilon_3}{s \frac{\theta}{2}} \cdot s\theta \\ &= \frac{\varepsilon_1 \varepsilon_2}{s^2 \frac{\theta}{2}} \cdot (1 - c\theta) + \frac{\varepsilon_3}{s \frac{\theta}{2}} \cdot s\theta \end{aligned}$$

By using trigonometric identities: $c\theta = 2c^2 \frac{\theta}{2} - 1$, and $s\theta = 2s \frac{\theta}{2} c \frac{\theta}{2}$, we have

$$\begin{aligned} k_x k_y v\theta + k_z s\theta &= \frac{\varepsilon_1 \varepsilon_2}{1 - c^2 \frac{\theta}{2}} \cdot \left(1 - 2c^2 \frac{\theta}{2} + 1\right) + \frac{\varepsilon_3}{s \frac{\theta}{2}} \cdot 2s \frac{\theta}{2} c \frac{\theta}{2} \\ &= \frac{2\varepsilon_1 \varepsilon_2}{1 - c^2 \frac{\theta}{2}} \cdot \left(1 - c^2 \frac{\theta}{2}\right) + 2\varepsilon_3 c \frac{\theta}{2} \\ &= 2\varepsilon_1 \varepsilon_2 + 2\varepsilon_3 \varepsilon_4 \end{aligned}$$

Proven!

[3 marks for calculation]

(c) When calculating the angle θ from the rotation matrix, it was determined that $\theta = 180^\circ$. As such, we cannot use the formulae in lecture notes to obtain k_x , k_y and k_z . Derive alternative expressions to calculate k_x , k_y and k_z in such a situation (i.e. when $\theta = 180^\circ$).

[5 marks]

When $\theta = 180^\circ$, we have $c\theta = -1$, $s\theta = 0$, $v\theta = 1 - c\theta = 2$. Putting these into the rotation matrix, we have:

$${}^A_B R = \begin{bmatrix} 2k_x^2 - 1 & 2k_x k_y & 2k_x k_z \\ 2k_x k_y & 2k_y^2 - 1 & 2k_y k_z \\ 2k_x k_z & 2k_y k_z & 2k_z^2 - 1 \end{bmatrix}$$

It is then easy to see that:

$$\begin{aligned} 2k_x^2 - 1 &= r_{11} \\ 2k_y^2 - 1 &= r_{22} \end{aligned}$$

$$2k_z^2 - 1 = r_{33}$$

As such:

$$k_x = \sqrt{\frac{r_{11} + 1}{2}}$$

$$k_y = \sqrt{\frac{r_{22} + 1}{2}}$$

$$k_z = \sqrt{\frac{r_{33} + 1}{2}}$$

[3 marks for reasoning, 2 marks for calculation]

(d) Show that the first column of the rotation matrix is of unit length.

[2 marks]

The magnitude of the first column can be calculated as:

$$\begin{aligned} \text{Magnitude} &= \sqrt{(k_x^2 v \theta + c \theta)^2 + (k_x k_y v \theta + k_z s \theta)^2 + (k_x k_z v \theta - k_y s \theta)^2} \\ &= \sqrt{\begin{aligned} &k_x^4 v^2 \theta + 2k_x^2 v \theta c \theta + c^2 \theta \\ &+ k_x^2 k_y^2 v^2 \theta + 2k_x k_y k_z v \theta s \theta + k_z^2 s^2 \theta \\ &+ k_x^2 k_z^2 v^2 \theta - 2k_x k_y k_z v \theta s \theta + k_y^2 s^2 \theta \end{aligned}} \\ &= \sqrt{\begin{aligned} &k_x^4 v^2 \theta + 2k_x^2 v \theta c \theta + c^2 \theta \\ &+ k_x^2 k_y^2 v^2 \theta + k_z^2 s^2 \theta \\ &+ k_x^2 k_z^2 v^2 \theta + k_y^2 s^2 \theta \end{aligned}} \\ &= \sqrt{k_x^2 \underbrace{(k_x^2 + k_y^2 + k_z^2)}_1 v^2 \theta + 2k_x^2 v \theta c \theta + c^2 \theta + k_z^2 s^2 \theta + k_y^2 s^2 \theta} \\ &= \sqrt{k_x^2 v^2 \theta + 2k_x^2 v \theta c \theta + c^2 \theta + k_z^2 s^2 \theta + k_y^2 s^2 \theta} \end{aligned}$$

Put all the trigonometric terms using cosine:

$$\begin{aligned} \text{Magnitude} &= \sqrt{k_x^2 (1 - c \theta)^2 + 2k_x^2 (1 - c \theta) c \theta + c^2 \theta + \underbrace{(k_y^2 + k_z^2)}_{1 - k_x^2} (1 - c^2 \theta)} \\ &= \sqrt{k_x^2 - 2k_x^2 c \theta + k_x^2 c^2 \theta + 2k_x^2 c \theta - 2k_x^2 c^2 \theta + c^2 \theta + 1 - k_x^2 - c^2 \theta + k_x^2 c^2 \theta} \\ &= 1 \end{aligned}$$

Proven!

[2 marks for calculation]

2. The human wrist has 3 degrees of freedom, as shown in Figures 2.1, 2.2 and 2.3.

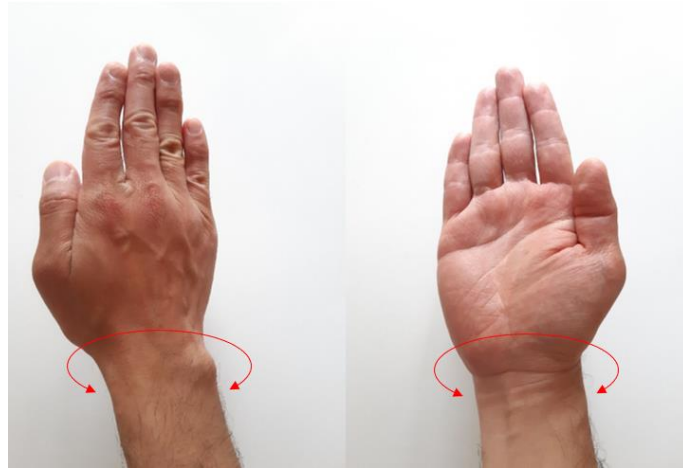


Figure 2.1: 1st degree of freedom – Pronation / Supination

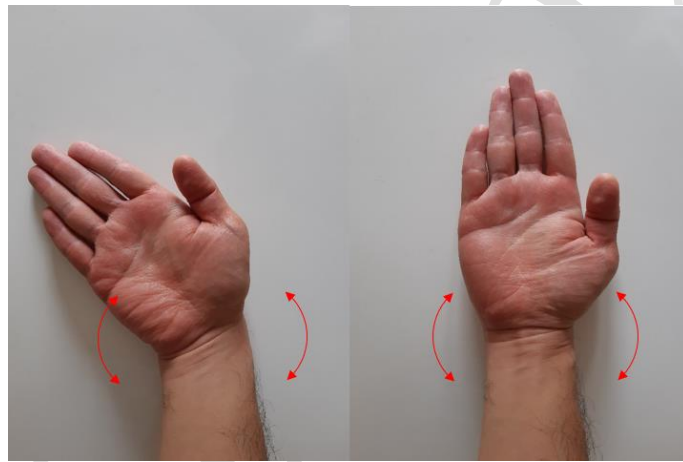


Figure 2.2: 2nd degree of freedom – Deviation

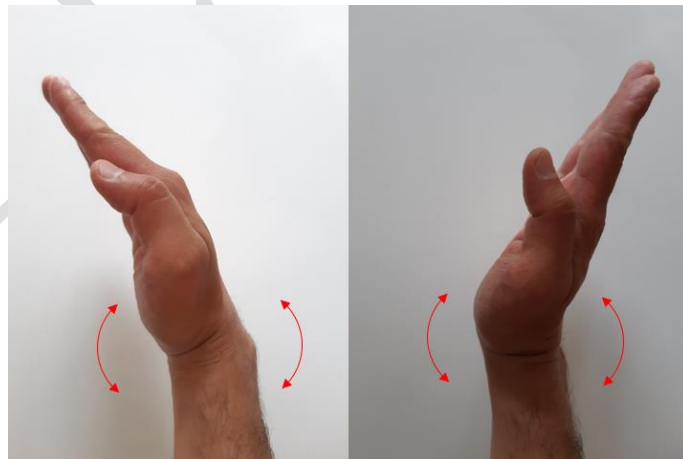


Figure 2.3: 3rd degree of freedom: Flexion / Extension

- (a) Derive the DH parameters of the wrist, where the order of the motions is first pronation / supination, followed by deviation, and ending with flexion / extension. The x_0 axis is given, and the configuration as shown in Figure 2.4 is when $\theta_1 = 0^\circ$. Show that the transformation matrix from frame {0} to frame {3} is:

$${}^0_3T = \begin{bmatrix} c\theta_1 c\theta_2 c\theta_3 + s\theta_1 s\theta_3 & -c\theta_1 c\theta_2 s\theta_3 + s\theta_1 c\theta_3 & c\theta_1 s\theta_2 & 0 \\ s\theta_1 c\theta_2 c\theta_3 - c\theta_1 s\theta_3 & -s\theta_1 c\theta_2 s\theta_3 - c\theta_1 c\theta_3 & s\theta_1 s\theta_2 & 0 \\ s\theta_2 c\theta_3 & -s\theta_2 s\theta_3 & -c\theta_2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

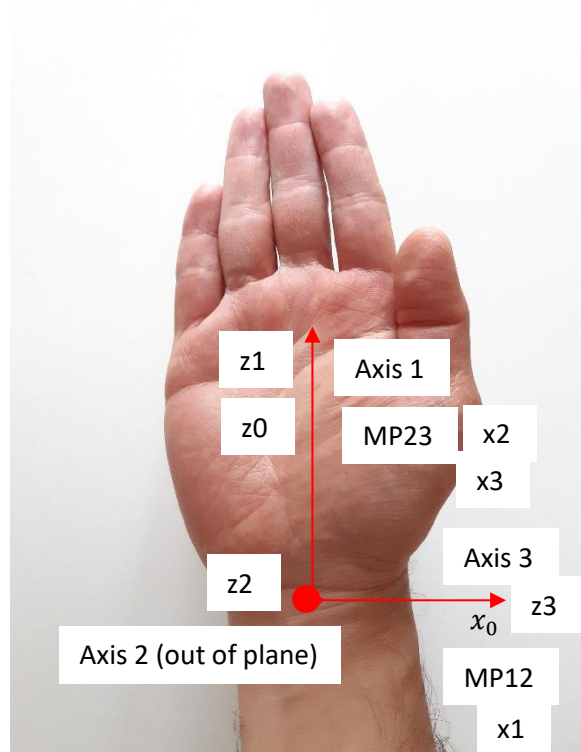


Figure 2.4: Initial configuration of wrist

Note:

- If there is any ambiguity in up-down direction, choose up.
- If there is any ambiguity in left-right direction, choose right.
- If there is any ambiguity in into the plane / out of the plane, choose out of the plane.

[8 marks]

Step 1: Draw axes (directly on Figure 2.4).

Step 2: Draw mutual perpendicular (MP) from lower axis to higher axis. Note: MP from axis 1 to axis 2 could be towards left or right. However, since x_0 is given, and since $\theta_1 = 0^\circ$ in the configuration as shown, MP_{12} must be the same as x_0 .

Step 3: Draw frames {1} and {2}. Z axis aligned with rotary/translation, where X axis aligned with mutual perpendiculars.

Step 4: Draw frame {0}. In this case, x_0 is already given. We only need to draw z_0 aligned with z_1 .

Step 5: Draw frame {3}. Z axis aligned with axis 3, X axis aligns with x_2 .

Step 6: Determine link lengths a_{i-1} :

a_0 = distance from z_0 to z_1 along $x_0 = 0$.

a_1 = distance from z_1 to z_2 along $x_1 = 0$.

a_2 = distance from z_2 to z_3 along $x_2 = 0$.

Step 7: Determine link twists α_{i-1} :

α_0 = angle between z_0 and z_1 about $x_0 = 0$.

α_1 = angle between z_1 and z_2 about $x_1 = 90^\circ$.

α_2 = angle between z_2 and z_3 about $x_2 = 90^\circ$.

Step 8: Determine link offsets d_i :

d_1 = distance from x_0 to x_1 along $z_1 = 0$.

d_2 = distance from x_1 to x_2 along $z_2 = 0$.

d_3 = distance from x_2 to x_3 along $z_3 = 0$.

Step 9: Determine joint angles θ_i :

θ_1 = angle between x_0 and x_1 about z_1 = variable.

θ_2 = angle between x_1 and x_2 about z_2 = variable.

θ_3 = angle between x_2 and x_3 about z_3 = variable.

Step 10: Summarize into a DH Table:

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	90	0	0	θ_2
3	90	0	0	θ_3

Step 11: Calculate the transformation matrices between frames based on:

$${}^{i-1}_iT = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ c\alpha_{i-1}s\theta_i & c\alpha_{i-1}c\theta_i & -s\alpha_{i-1} & -s\alpha_{i-1}d_i \\ s\alpha_{i-1}s\theta_i & s\alpha_{i-1}c\theta_i & c\alpha_{i-1} & c\alpha_{i-1}d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^0_1T = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & 0 \\ s\theta_1 & c\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, {}^1_2T = \begin{bmatrix} c\theta_2 & -s\theta_2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ s\theta_2 & c\theta_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, {}^2_3T = \begin{bmatrix} c\theta_3 & -s\theta_3 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ s\theta_3 & c\theta_3 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{aligned} {}^0_3T &= {}^0_1T \cdot {}^1_2T \cdot {}^2_3T = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & 0 \\ s\theta_1 & c\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} c\theta_2 & -s\theta_2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ s\theta_2 & c\theta_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} c\theta_3 & -s\theta_3 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ s\theta_3 & c\theta_3 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} c\theta_1c\theta_2 & -c\theta_1s\theta_2 & s\theta_1 & 0 \\ s\theta_1c\theta_2 & -s\theta_1s\theta_2 & -c\theta_1 & 0 \\ s\theta_2 & c\theta_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} c\theta_3 & -s\theta_3 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ s\theta_3 & c\theta_3 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} c\theta_1c\theta_2c\theta_3 + s\theta_1s\theta_3 & -c\theta_1c\theta_2s\theta_3 + s\theta_1c\theta_3 & c\theta_1s\theta_2 & 0 \\ s\theta_1c\theta_2c\theta_3 - c\theta_1s\theta_3 & -s\theta_1c\theta_2s\theta_3 - c\theta_1c\theta_3 & s\theta_1s\theta_2 & 0 \\ s\theta_2c\theta_3 & -s\theta_2s\theta_3 & -c\theta_2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

[2 marks for steps 1-5; 1 marks each for steps 6-9; 2 marks for step 10-11].

- (b) Derive the rotational Jacobian matrix of the wrist, with respect to the wrist's frame {0}. Does the wrist have any rotational singularity?

Note: If you do not have the intermediate transformation matrices in part (a), or if you are unsure of your answers in part (a), please use the following for calculation of part (b):

$${}^0_1T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & c\theta_1 & -s\theta_1 & 0 \\ 0 & s\theta_1 & c\theta_1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, {}^1_2T = \begin{bmatrix} c\theta_2 & 0 & s\theta_2 & 0 \\ 0 & 1 & 0 & 0 \\ -s\theta_2 & 0 & c\theta_2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, {}^2_3T = \begin{bmatrix} c\theta_3 & -s\theta_3 & 0 & 0 \\ s\theta_3 & c\theta_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

[12 marks]

Using the velocity propagation method, we start with ${}^0\omega_0 = 0$.

Then:

$${}^1\omega_1 = {}^0R \cdot \underbrace{{}^0\omega_0}_0 + \dot{\theta}_1 \cdot {}^1\hat{z}_1 = \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix}$$

$${}^2\omega_2 = \underbrace{{}^1R}_{{}^1_2R^T} \cdot {}^1\omega_1 + \dot{\theta}_2 \cdot {}^2\hat{z}_2 = \begin{bmatrix} c\theta_2 & 0 & s\theta_2 \\ -s\theta_2 & 0 & c\theta_2 \\ 0 & -1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_2 \end{bmatrix} = \begin{bmatrix} s\theta_2 \cdot \dot{\theta}_1 \\ c\theta_2 \cdot \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}$$

$$\begin{aligned} {}^3\omega_3 &= \underbrace{{}^2R}_{{}^2_3R^T} \cdot {}^2\omega_2 + \dot{\theta}_3 \cdot {}^3\hat{z}_3 = \begin{bmatrix} c\theta_3 & 0 & s\theta_3 \\ -s\theta_3 & 0 & c\theta_3 \\ 0 & -1 & 0 \end{bmatrix} \cdot \begin{bmatrix} s\theta_2 \cdot \dot{\theta}_1 \\ c\theta_2 \cdot \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_3 \end{bmatrix} \\ &= \begin{bmatrix} s\theta_2 \cdot c\theta_3 \cdot \dot{\theta}_1 + s\theta_3 \cdot \dot{\theta}_2 \\ -s\theta_2 \cdot s\theta_3 \cdot \dot{\theta}_1 + c\theta_3 \cdot \dot{\theta}_2 \\ -c\theta_2 \cdot \dot{\theta}_1 + \dot{\theta}_3 \end{bmatrix} \end{aligned}$$

We now transform the result back to the base frame:

$$\begin{aligned} {}^0\omega_3 &= {}^0R \cdot {}^3\omega_3 \\ &= \begin{bmatrix} c\theta_1 c\theta_2 c\theta_3 + s\theta_1 s\theta_3 & -c\theta_1 c\theta_2 s\theta_3 + s\theta_1 c\theta_3 & c\theta_1 s\theta_2 \\ s\theta_1 c\theta_2 c\theta_3 - c\theta_1 s\theta_3 & -s\theta_1 c\theta_2 s\theta_3 - c\theta_1 c\theta_3 & s\theta_1 s\theta_2 \\ s\theta_2 c\theta_3 & -s\theta_2 s\theta_3 & -c\theta_2 \end{bmatrix} \begin{bmatrix} s\theta_2 \cdot c\theta_3 \cdot \dot{\theta}_1 + s\theta_3 \cdot \dot{\theta}_2 \\ -s\theta_2 \cdot s\theta_3 \cdot \dot{\theta}_1 + c\theta_3 \cdot \dot{\theta}_2 \\ -c\theta_2 \cdot \dot{\theta}_1 + \dot{\theta}_3 \end{bmatrix} \\ &= \begin{bmatrix} (c\theta_1 c\theta_2 c\theta_3 + s\theta_1 s\theta_3)(s\theta_2 c\theta_3 \dot{\theta}_1 + s\theta_3 \dot{\theta}_2) + (-c\theta_1 c\theta_2 s\theta_3 + s\theta_1 c\theta_3)(-s\theta_2 s\theta_3 \dot{\theta}_1 + c\theta_3 \dot{\theta}_2) + c\theta_1 s\theta_2(-c\theta_2 \dot{\theta}_1 + \dot{\theta}_3) \\ (s\theta_1 c\theta_2 c\theta_3 - c\theta_1 s\theta_3)(s\theta_2 c\theta_3 \dot{\theta}_1 + s\theta_3 \dot{\theta}_2) + (-s\theta_1 c\theta_2 s\theta_3 - c\theta_1 c\theta_3)(-s\theta_2 s\theta_3 \dot{\theta}_1 + c\theta_3 \dot{\theta}_2) + s\theta_1 s\theta_2(-c\theta_2 \dot{\theta}_1 + \dot{\theta}_3) \\ s\theta_2 c\theta_3(s\theta_2 c\theta_3 \dot{\theta}_1 + s\theta_3 \dot{\theta}_2) - s\theta_2 s\theta_3(-s\theta_2 s\theta_3 \dot{\theta}_1 + c\theta_3 \dot{\theta}_2) - c\theta_2(-c\theta_2 \dot{\theta}_1 + \dot{\theta}_3) \end{bmatrix} \\ &= \begin{bmatrix} (c\theta_1 c\theta_2 c\theta_3 + s\theta_1 s\theta_3)s\theta_2 c\theta_3 & (c\theta_1 c\theta_2 c\theta_3 + s\theta_1 s\theta_3)s\theta_3 & c\theta_1 s\theta_2 \\ -(-c\theta_1 c\theta_2 s\theta_3 + s\theta_1 c\theta_3)s\theta_2 s\theta_3 - c\theta_1 s\theta_2 c\theta_2 & +(-c\theta_1 c\theta_2 s\theta_3 + s\theta_1 c\theta_3)c\theta_3 & \\ (s\theta_1 c\theta_2 c\theta_3 - c\theta_1 s\theta_3)s\theta_2 c\theta_3 & (s\theta_1 c\theta_2 c\theta_3 - c\theta_1 s\theta_3)s\theta_3 & s\theta_1 s\theta_2 \\ -(-s\theta_1 c\theta_2 s\theta_3 - c\theta_1 c\theta_3)s\theta_2 s\theta_3 - s\theta_1 s\theta_2 c\theta_2 & +(-s\theta_1 c\theta_2 s\theta_3 - c\theta_1 c\theta_3)c\theta_3 & \\ s\theta_2 c\theta_3 s\theta_2 c\theta_3 + s\theta_2 s\theta_3 s\theta_2 s\theta_3 + c\theta_2 c\theta_2 & s\theta_2 c\theta_3 s\theta_3 - s\theta_2 s\theta_3 c\theta_3 & -c\theta_2 \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \end{bmatrix} \end{aligned}$$

Therefore, the rotational Jacobian matrix is:

$$\begin{aligned}
& \begin{bmatrix} (c\theta_1 c\theta_2 c\theta_3 + s\theta_1 s\theta_3) s\theta_2 c\theta_3 & (c\theta_1 c\theta_2 c\theta_3 + s\theta_1 s\theta_3) s\theta_3 & c\theta_1 s\theta_2 \\ -(-c\theta_1 c\theta_2 s\theta_3 + s\theta_1 c\theta_3) s\theta_2 s\theta_3 - c\theta_1 s\theta_2 c\theta_2 & +(-c\theta_1 c\theta_2 s\theta_3 + s\theta_1 c\theta_3) c\theta_3 & \\ (s\theta_1 c\theta_2 c\theta_3 - c\theta_1 s\theta_3) s\theta_2 c\theta_3 & (s\theta_1 c\theta_2 c\theta_3 - c\theta_1 s\theta_3) s\theta_3 & s\theta_1 s\theta_2 \\ -(-s\theta_1 c\theta_2 s\theta_3 - c\theta_1 c\theta_3) s\theta_2 s\theta_3 - s\theta_1 s\theta_2 c\theta_2 & +(-s\theta_1 c\theta_2 s\theta_3 - c\theta_1 c\theta_3) c\theta_3 & \\ s\theta_2 c\theta_3 s\theta_2 c\theta_3 + s\theta_2 s\theta_3 s\theta_2 s\theta_3 + c\theta_2 c\theta_2 & s\theta_2 c\theta_3 s\theta_3 - s\theta_2 s\theta_3 c\theta_3 & -c\theta_2 \end{bmatrix} \\
&= \begin{bmatrix} c\theta_1 c\theta_2 c\theta_3 s\theta_2 c\theta_3 + s\theta_1 s\theta_3 s\theta_2 c\theta_3 & c\theta_1 c\theta_2 c\theta_3 s\theta_3 + s\theta_1 s\theta_3 s\theta_3 & c\theta_1 s\theta_2 \\ +c\theta_1 c\theta_2 s\theta_3 s\theta_2 s\theta_3 - s\theta_1 c\theta_3 s\theta_2 s\theta_3 - c\theta_1 s\theta_2 c\theta_2 & -c\theta_1 c\theta_2 s\theta_3 c\theta_3 + s\theta_1 c\theta_3 c\theta_3 & \\ s\theta_1 c\theta_2 c\theta_3 s\theta_2 c\theta_3 - c\theta_1 s\theta_3 s\theta_2 c\theta_3 & s\theta_1 c\theta_2 c\theta_3 s\theta_3 - c\theta_1 s\theta_3 s\theta_3 & s\theta_1 s\theta_2 \\ +s\theta_1 c\theta_2 s\theta_3 s\theta_2 s\theta_3 + c\theta_1 c\theta_3 s\theta_2 s\theta_3 - s\theta_1 s\theta_2 c\theta_2 & -s\theta_1 c\theta_2 s\theta_3 c\theta_3 - c\theta_1 c\theta_3 c\theta_3 & \\ s\theta_2 c\theta_3 s\theta_2 c\theta_3 + s\theta_2 s\theta_3 s\theta_2 s\theta_3 + c\theta_2 c\theta_2 & s\theta_2 c\theta_3 s\theta_3 - s\theta_2 s\theta_3 c\theta_3 & -c\theta_2 \end{bmatrix} \\
&= \begin{bmatrix} 0 & s\theta_1 & c\theta_1 s\theta_2 \\ 0 & -c\theta_1 & s\theta_1 s\theta_2 \\ 1 & 0 & -c\theta_2 \end{bmatrix}
\end{aligned}$$

To calculate singularity, we check the determinant of the Jacobian matrix:

$$\begin{vmatrix} 0 & s\theta_1 & c\theta_1 s\theta_2 \\ 0 & -c\theta_1 & s\theta_1 s\theta_2 \\ 1 & 0 & -c\theta_2 \end{vmatrix} = 1 \cdot \begin{vmatrix} s\theta_1 & c\theta_1 s\theta_2 \\ -c\theta_1 & s\theta_1 s\theta_2 \end{vmatrix} = s\theta_1 s\theta_1 s\theta_2 + c\theta_1 c\theta_1 s\theta_2 = s\theta_2$$

Yes, the wrist has rotational singularity when $s\theta_2 = 0$, or $\theta_2 = 0, \pm k\pi$.

[3 marks for calculating the 3 velocities, 2 marks for transforming back to base frame, 2 marks for putting into Jacobian form, 2 marks for simplification, 3 marks for singularity].

3. Calculate the mass moments of inertia, I_{xx} and I_{yy} (with respect to the centre of mass) of a solid long rod of length l , with an equilateral triangular cross-sectional area, as shown in Figure 3.1. The mass of the rod is m , and the density is ρ .

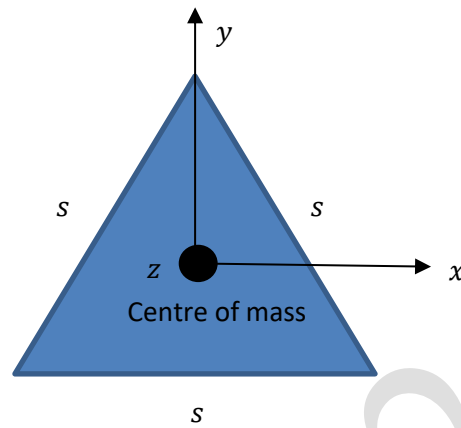
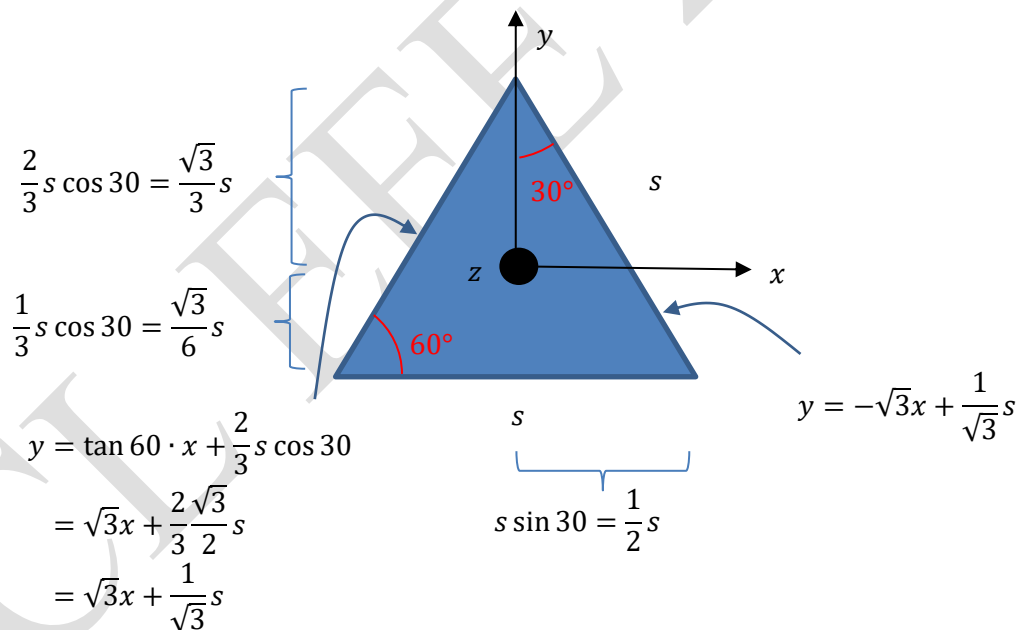


Figure 3.1: Equilateral triangle

[20 marks]

First, we put in the dimensions for the triangle:



Now, x and y are related. For each level of y , the limits of x are from $x = \frac{1}{\sqrt{3}}\left(y - \frac{1}{\sqrt{3}}s\right)$ to $x = -\frac{1}{\sqrt{3}}\left(y - \frac{1}{\sqrt{3}}s\right)$.

Therefore:

$$\begin{aligned}
I_{xx} &= \iiint_V (y^2 + z^2) \rho dV \\
&= \int_{z=-\frac{l}{2}}^{\frac{l}{2}} \int_{y=-\frac{\sqrt{3}}{6}s}^{\frac{\sqrt{3}}{3}s} \int_{x=\frac{1}{\sqrt{3}}(y-\frac{1}{\sqrt{3}}s)}^{-\frac{1}{\sqrt{3}}(y-\frac{1}{\sqrt{3}}s)} (y^2 + z^2) \rho \cdot dx \cdot dy \cdot dz \\
&= \int_{z=-\frac{l}{2}}^{\frac{l}{2}} \int_{y=-\frac{\sqrt{3}}{6}s}^{\frac{\sqrt{3}}{3}s} (y^2 + z^2) x \Big|_{x=\frac{1}{\sqrt{3}}(y-\frac{1}{\sqrt{3}}s)}^{-\frac{1}{\sqrt{3}}(y-\frac{1}{\sqrt{3}}s)} \rho \cdot dy \cdot dz \\
&= -\frac{2}{\sqrt{3}} \int_{z=-\frac{l}{2}}^{\frac{l}{2}} \int_{y=-\frac{\sqrt{3}}{6}s}^{\frac{\sqrt{3}}{3}s} (y^2 + z^2) \left(y - \frac{1}{\sqrt{3}}s \right) \rho \cdot dy \cdot dz \\
&= -\frac{2}{\sqrt{3}} \int_{z=-\frac{l}{2}}^{\frac{l}{2}} \int_{y=-\frac{\sqrt{3}}{6}s}^{\frac{\sqrt{3}}{3}s} \left(y^3 - \frac{1}{\sqrt{3}}sy^2 + z^2y - \frac{1}{\sqrt{3}}z^2s \right) \rho \cdot dy \cdot dz \\
&= -\frac{2}{\sqrt{3}} \int_{z=-\frac{l}{2}}^{\frac{l}{2}} \left(\frac{y^4}{4} - \frac{1}{3\sqrt{3}}sy^3 + \frac{z^2y^2}{2} - \frac{1}{\sqrt{3}}z^2sy \right) \Big|_{y=-\frac{\sqrt{3}}{6}s}^{\frac{\sqrt{3}}{3}s} \rho \cdot dz \\
&= -\frac{2}{\sqrt{3}} \int_{z=-\frac{l}{2}}^{\frac{l}{2}} \left(\frac{s^4}{36} - \frac{s^4}{27} + \frac{z^2s^2}{6} - \frac{z^2s^2}{3} - \frac{s^4}{576} - \frac{s^4}{216} - \frac{z^2s^2}{24} - \frac{z^2s^2}{6} \right) \rho \cdot dz \\
&= -\frac{2}{\sqrt{3}} \int_{z=-\frac{l}{2}}^{\frac{l}{2}} \left(\frac{48 - 64 - 3 - 8}{1728} s^4 + \frac{4 - 8 - 1 - 4}{24} z^2s^2 \right) \rho \cdot dz \\
&= \frac{2}{\sqrt{3}} \int_{z=-\frac{l}{2}}^{\frac{l}{2}} \left(\frac{1}{64} s^4 + \frac{3}{8} z^2s^2 \right) \rho \cdot dz \\
&= \frac{2}{\sqrt{3}} \left(\frac{1}{64} s^4 z + \frac{1}{8} z^3 s^2 \right) \Big|_{z=-\frac{l}{2}}^{\frac{l}{2}} \rho \\
&= \frac{2}{\sqrt{3}} \left(\frac{1}{128} s^4 l + \frac{1}{64} l^3 s^2 + \frac{1}{128} s^4 l + \frac{1}{64} l^3 s^2 \right) \rho \\
&= \frac{2}{\sqrt{3}} \left(\frac{1}{64} s^4 l + \frac{1}{32} l^3 s^2 \right) \rho \\
&= \left(\frac{1}{32\sqrt{3}} s^4 l + \frac{1}{16\sqrt{3}} l^3 s^2 \right) \rho \\
&= \frac{\sqrt{3}}{4} s^2 l \rho \left(\frac{1}{24} s^2 + \frac{1}{12} l^2 \right) \\
&\quad \underbrace{\quad}_{\substack{\text{area} \\ \text{volume} \\ \text{mass}}} \\
&= m \left(\frac{1}{24} s^2 + \frac{1}{12} l^2 \right)
\end{aligned}$$

Next, we have:

$$\begin{aligned}
I_{yy} &= \iiint_V (x^2 + z^2) \rho dV \\
&= \int_{z=-\frac{l}{2}}^{\frac{l}{2}} \int_{y=-\frac{\sqrt{3}}{6}s}^{\frac{\sqrt{3}}{3}s} \int_{x=\frac{1}{\sqrt{3}}(y-\frac{1}{\sqrt{3}}s)}^{-\frac{1}{\sqrt{3}}(y-\frac{1}{\sqrt{3}}s)} (x^2 + z^2) \rho \cdot dx \cdot dy \cdot dz
\end{aligned}$$

$$\begin{aligned}
&= \int_{z=-\frac{l}{2}}^{\frac{l}{2}} \int_{y=-\frac{\sqrt{3}}{6}s}^{\frac{\sqrt{3}}{3}s} \left(\frac{x^3}{3} + z^2 x \right) \bigg|_{x=\frac{1}{\sqrt{3}}\left(y-\frac{1}{\sqrt{3}}s\right)}^{-\frac{1}{\sqrt{3}}\left(y-\frac{1}{\sqrt{3}}s\right)} \rho \cdot dy \cdot dz \\
&= \int_{z=-\frac{l}{2}}^{\frac{l}{2}} \int_{y=-\frac{\sqrt{3}}{6}s}^{\frac{\sqrt{3}}{3}s} \left(-\frac{2}{9\sqrt{3}}y^3 + \frac{2}{9}y^2s - \frac{2}{9\sqrt{3}}ys^2 + \frac{2}{81}s^3 - \frac{2}{\sqrt{3}}z^2y + \frac{2}{3}z^2s \right) \rho \cdot dy \cdot dz \\
&= \int_{z=-\frac{l}{2}}^{\frac{l}{2}} \left(-\frac{1}{18\sqrt{3}}y^4 + \frac{2}{27}y^3s - \frac{1}{9\sqrt{3}}y^2s^2 + \frac{2}{81}s^3y - \frac{1}{\sqrt{3}}z^2y^2 + \frac{2}{3}z^2sy \right) \bigg|_{y=-\frac{\sqrt{3}}{6}s}^{\frac{\sqrt{3}}{3}s} \rho \cdot dz \\
&= \int_{z=-\frac{l}{2}}^{\frac{l}{2}} \left(\frac{\sqrt{3}}{96}s^4 + \frac{\sqrt{3}}{4}z^2s^2 \right) \rho \cdot dz \\
&= \left(\frac{\sqrt{3}}{96}s^4z + \frac{\sqrt{3}}{12}z^3s^2 \right) \bigg|_{z=-\frac{l}{2}}^{\frac{l}{2}} \rho \\
&= \left(\frac{\sqrt{3}}{96}s^4l + \frac{\sqrt{3}}{48}s^2l^3 \right) \rho \\
&= \underbrace{\frac{\sqrt{3}}{4}s^2l}_{\substack{\text{area} \\ \text{volume} \\ \text{mass}}} \rho \left(\frac{1}{24}s^2 + \frac{1}{12}l^2 \right) \\
&= m \left(\frac{1}{24}s^2 + \frac{1}{12}l^2 \right)
\end{aligned}$$

[5 marks for putting in dimensions into triangle and getting line equations, 2 mark for getting the limits of integration, 5 marks for I_{xx} up until before changing to volume and mass, 2 mark for changing to volume and mass, 5 marks for I_{yy} up until before changing to volume and mass, 1 mark for changing to volume and mass]

4. We want to move a robot's joint from θ_0 to θ_f in time t_f along a cubic trajectory, with zero velocities at the starting position and final position. Assume $\theta_f > \theta_0$.

(a) The values of θ_0 and θ_f are known. What should t_f be, if the velocity and acceleration cannot exceed $\dot{\theta}_{max}$ and $\ddot{\theta}_{max}$ respectively?

[10 marks]

The cubic trajectory is:

$$\theta(t) = \theta_0 + \frac{3}{t_f^2}(\theta_f - \theta_0)t^2 - \frac{2}{t_f^3}(\theta_f - \theta_0)t^3$$

The velocity is thus:

$$\dot{\theta}(t) = \frac{6}{t_f^2}(\theta_f - \theta_0)t - \frac{6}{t_f^3}(\theta_f - \theta_0)t^2$$

Whereas the acceleration is:

$$\ddot{\theta}(t) = \frac{6}{t_f^2}(\theta_f - \theta_0) - \frac{12}{t_f^3}(\theta_f - \theta_0)t$$

The acceleration graph is a straight line, starting at $\frac{6}{t_f^2}(\theta_f - \theta_0)$ at $t = 0$, and decreases with time because of the negative gradient. As such, the maximum of acceleration for a cubic polynomial occurs at time $t = 0$, and we require this to be less than $\ddot{\theta}_{max}$. Thus:

$$\ddot{\theta}(t = 0) = \frac{6}{t_f^2}(\theta_f - \theta_0) \leq \ddot{\theta}_{max}$$

This gives:

$$t_f^2 \geq \frac{6}{\ddot{\theta}_{max}}(\theta_f - \theta_0)$$

Thus, t_f based on maximum acceleration is:

$$t_f \geq \sqrt{\frac{6}{\ddot{\theta}_{max}}(\theta_f - \theta_0)}$$

As for velocity, the maximum value happens at $t = \frac{t_f}{2}$, which can be shown by sketching a graph of velocity, or setting the derivative namely the acceleration as zero ($\ddot{\theta}(t) = \frac{6}{t_f^2}(\theta_f - \theta_0) - \frac{12}{t_f^3}(\theta_f - \theta_0)t = 0 \rightarrow \frac{12}{t_f^3}(\theta_f - \theta_0)t = \frac{6}{t_f^2}(\theta_f - \theta_0) \rightarrow t = \frac{t_f}{2}$). We require this to be less than $\dot{\theta}_{max}$, thus:

$$\dot{\theta}\left(t = \frac{t_f}{2}\right) = \frac{6}{t_f^2}(\theta_f - \theta_0)\frac{t_f}{2} - \frac{6}{t_f^3}(\theta_f - \theta_0)\frac{t_f^2}{4} \leq \dot{\theta}_{max}$$

$$\frac{3}{t_f}(\theta_f - \theta_0) - \frac{3}{2t_f}(\theta_f - \theta_0) \leq \dot{\theta}_{max}$$

$$\frac{3}{2t_f}(\theta_f - \theta_0) \leq \dot{\theta}_{max}$$

Thus, t_f based on maximum velocity is:

$$t_f \geq \frac{3}{2\dot{\theta}_{max}}(\theta_f - \theta_0)$$

Finally, we need to choose the larger of the two:

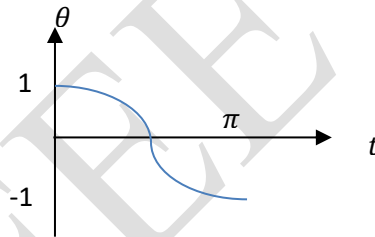
$$t_f \geq \max\left(\frac{3}{2\dot{\theta}_{max}}(\theta_f - \theta_0), \sqrt{\frac{6}{\ddot{\theta}_{max}}(\theta_f - \theta_0)}\right)$$

[4 marks for getting value based on velocity, 4 marks for getting value based on acceleration, and 2 mark for stating that the value needs to be the higher of the two]

- (b) You are a creative student and decided to design your own trajectory rather than the ones you learnt in the class. One idea you have is to use the half cycle of a cosine function as a trajectory. Write down the equation of such a trajectory, parameterized by θ_0 , θ_f and t_f . Please show how you arrive at the equation, e.g. by sketches of the curves.

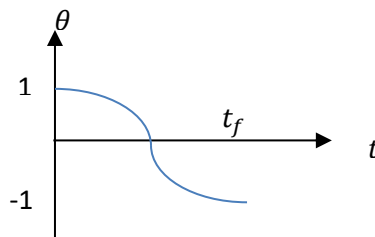
[10 marks]

The first half cycle of a cosine function ($\theta = \cos t$) is:



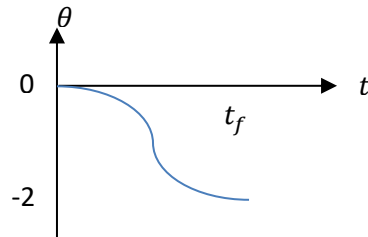
We can first “normalize” the time axis, such that the curve ends at $t = t_f$:

$$\theta = \cos\left(\frac{\pi}{t_f}t\right)$$



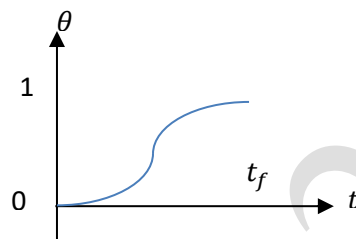
Next, we shift the curve downwards by 1, so that it starts at $\theta = 0$:

$$\theta = \cos\left(\frac{\pi}{t_f}t\right) - 1$$



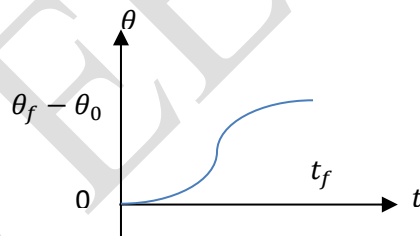
Multiply this by -0.5 to flip the curve into positive direction, and with a final x of 1.

$$\theta = -\frac{1}{2}\left(\cos\left(\frac{\pi}{t_f}t\right) - 1\right)$$



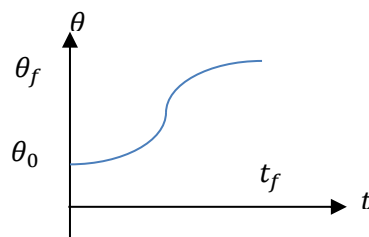
The trajectory should start from θ_0 to θ_f . For the moment, let it start from 0. Thus, the final position will be $\theta_f - \theta_0$. This can be done by multiplying the whole term by $\theta_f - \theta_0$:

$$\theta = -\frac{1}{2}(\theta_f - \theta_0)\left(\cos\left(\frac{\pi}{t_f}t\right) - 1\right)$$



Finally, we add a constant θ_0 to arrive at the final equation:

$$\theta = \theta_0 - \frac{1}{2}(\theta_f - \theta_0)\left(\cos\left(\frac{\pi}{t_f}t\right) - 1\right)$$



[There are 5 “steps” in the derivation above after the initial cosine expression. 2 marks given for each step.]

UCL FEE 2021-22

5.

(a) For a single joint of the manipulator, the dynamics of the load is:

$$(I + I_m \eta^2) \ddot{\theta} + (b + b_m \eta^2) \dot{\theta} = \tau - \tau_{\text{fric}}$$

where $I = 10$, $I_m = 1$, $b = 5$, $b_m = 1$, $\eta = 5$, and τ_{fric} is a viscous friction with gradient $k = 0.2$.

Derive a controller such that the single joint can track a desired trajectory $(\theta_d, \dot{\theta}_d, \ddot{\theta}_d)$ with a closed-loop stiffness of 5 and closed-loop damping of 4. Sketch the implementation of the controller (e.g. sensor input, control output).

[6 marks]

System is:

$$(I + I_m \eta^2) \ddot{\theta} + (b + b_m \eta^2) \dot{\theta} = \tau - \tau_{\text{fric}}$$

Rearranged:

$$(I + I_m \eta^2) \ddot{\theta} + (b + b_m \eta^2) \dot{\theta} + \tau_{\text{fric}} = \tau$$

Putting in the viscous friction:

$$(I + I_m \eta^2) \ddot{\theta} + (b + b_m \eta^2) \dot{\theta} + k \dot{\theta} = \tau$$

Rearranged:

$$(I + I_m \eta^2) \ddot{\theta} + (b + b_m \eta^2 + k) \dot{\theta} = \tau$$

Now, we can design the controller according to lecture notes. First, we design the model-compensator:

$$\tau = (I + I_m \eta^2) f + (b + b_m \eta^2 + k) \dot{\theta}$$

The closed loop equation then becomes:

$$\begin{aligned} (I + I_m \eta^2) \ddot{\theta} + (b + b_m \eta^2 + k) \dot{\theta} &= \tau \\ &= (I + I_m \eta^2) f + (b + b_m \eta^2 + k) \dot{\theta} \end{aligned}$$

which after cancellation of similar terms results in:

$$\ddot{\theta} = f$$

Then, we design the servo controller for trajectory tracking:

$$f = \ddot{\theta}_d + k_v(\dot{\theta}_d - \dot{\theta}) + k_p(\theta_d - \theta)$$

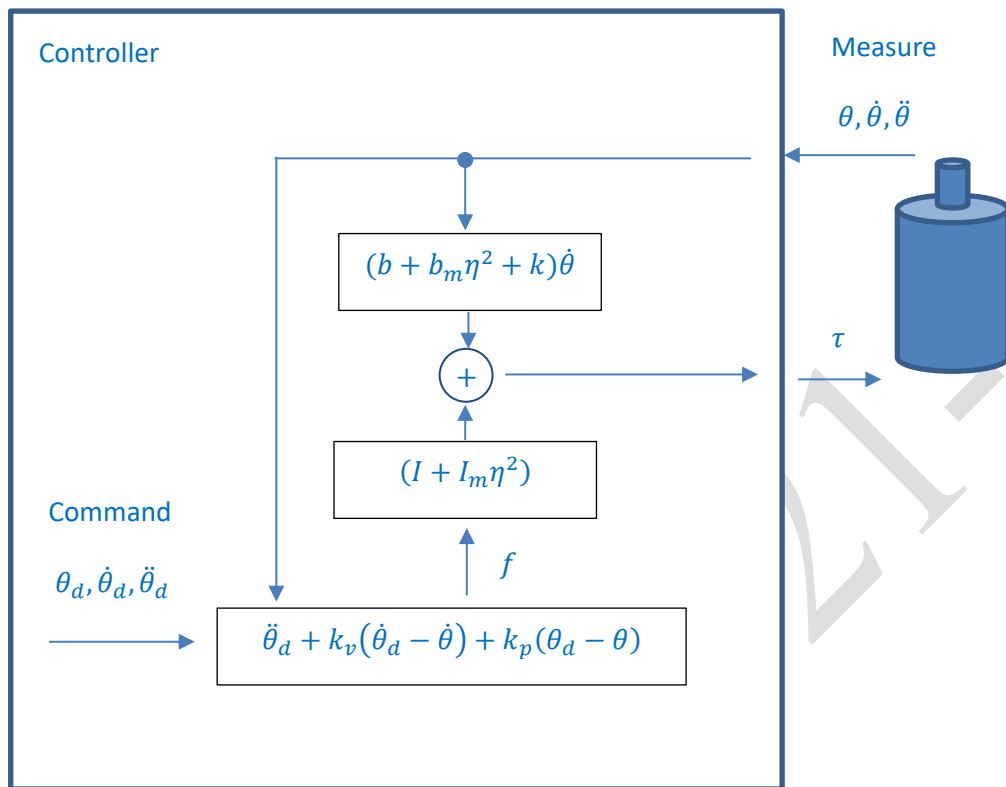
With this, the final closed loop equation becomes:

$$\ddot{\theta} = f = \ddot{\theta}_d + k_v(\dot{\theta}_d - \dot{\theta}) + k_p(\theta_d - \theta)$$

$$(\ddot{\theta}_d - \ddot{\theta}) + k_v(\dot{\theta}_d - \dot{\theta}) + k_p(\theta_d - \theta) = 0$$

To achieve closed-loop stiffness of 5 and closed-loop damping of 4, set $k_p = 5$, $k_v = 4$.

Controller implementation:



[3 marks for controller derivation, 3 marks for sketch]

- (b) Assume that the controller in part (a) has been designed correctly. Determine the equation of the tracking error $e = \theta_d - \theta$, if the initial tracking error is 4, and the initial rate of change of tracking error is 0. Sketch a time-plot of the tracking error, as well as the motion of the joint if $\theta_d = 4$ (a constant).

[10 marks]

The closed loop equation in terms of tracking error is:

$$\ddot{e} + k_v \dot{e} + k_p e = 0$$

Putting in the numbers, we have:

$$\ddot{e} + 4\dot{e} + 5e = 0$$

The characteristic equation of the 2nd order differential equation is:

$$\lambda^2 + 4\lambda + 5 = 0$$

The roots of the characteristic equation is:

$$\begin{aligned} \lambda &= \frac{-4 \pm \sqrt{16 - 20}}{2} \\ &= \frac{-4 \pm j\sqrt{4}}{2} \\ &= -2 \pm j \end{aligned}$$

The general equation for the error is therefore:

$$e(t) = c_1 e^{-2t} \cos t + c_2 e^{-2t} \sin t$$

To find the values of c_1 and c_2 , we use the initial conditions.

At $t = 0$,

$$e(t = 0) = c_1 = 4$$

The rate of change of error is:

$$\dot{e}(t) = -2c_1 e^{-2t} \cos t - c_1 e^{-2t} \sin t - 2c_2 e^{-2t} \sin t + c_2 e^{-2t} \cos t$$

At $t = 0$,

$$\dot{e}(t = 0) = -2c_1 + c_2 = 0$$

Thus:

$$c_2 = 2c_1 = 8$$

Hence:

$$\begin{aligned} e(t) &= 4e^{-2t} \cos t + 8e^{-2t} \sin t \\ &= 4e^{-2t} (\cos t + 2 \sin t) \\ &= 4e^{-2t} (A \sin(t + \varphi)) \\ &= 4e^{-2t} (A \sin t \cos \varphi + A \cos t \sin \varphi) \end{aligned}$$

Compare the coefficients:

$$A \cos \varphi = 2$$

$$A \sin \varphi = 1$$

Thus:

$$\tan \varphi = \frac{1}{2}$$

$$\varphi = \text{atan}(1/2)$$

Also,

$$A^2 = 2^2 + 1^2 = 5$$

$$A = \sqrt{5}$$

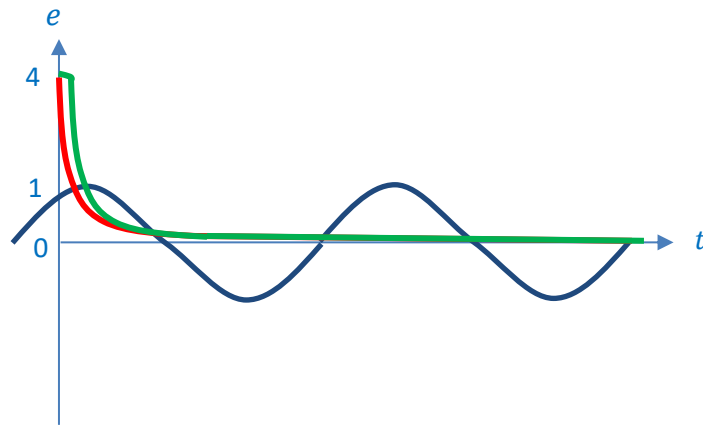
Therefore:

$$e(t) = 4\sqrt{5}e^{-2t} \sin(t + \text{atan}(1/2))$$

Sketch of error:

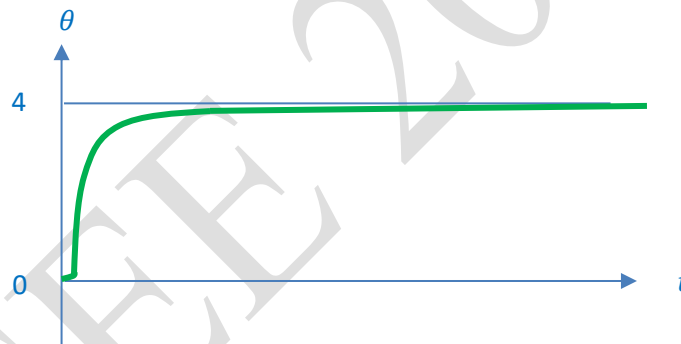
$$\sin(\text{atan } 1/2) = 1/\sqrt{5}$$

$$\sqrt{5} \sin(\text{atan } 1/2) = 1$$



— $\sqrt{5} \sin(t + \tan^{-1}(1/2))$
— $4e^{-2t}$
— product

Then, sketch of θ :



[2 marks for roots of characteristic equation, 2 marks for general solution, 2 marks for calculating c_1 and c_2 , 1 mark for putting cosine and sine into only one, 2 marks for sketch of error, 1 mark for sketch of theta]

- (c) Describe the behaviour of the θ response, if the closed-loop stiffness gets smaller and smaller.

[4 marks]

The closed loop equation in terms of tracking error is:

$$\ddot{e} + k_v \dot{e} + k_p e = 0$$

When stiffness is 5 and damping is 4, we have underdamped response. Now, consider the case when stiffness is 4. The equation becomes:

$$\ddot{e} + 4\dot{e} + 4e = 0$$

The characteristic equation becomes:

$$\lambda^2 + 4\lambda + 4 = 0$$

The roots of the characteristic equation is:

$$\lambda = \frac{-4 \pm \sqrt{16 - 16}}{2} = -2, -2$$

Because the roots are equal, we will have critically damped response.

If the stiffness reduces further, for e.g. to 3, the equation becomes:

$$\ddot{e} + 4\dot{e} + 3e = 0$$

The characteristic equation becomes:

$$\lambda^2 + 4\lambda + 3 = 0$$

The roots of the characteristic equation is:

$$\lambda = \frac{-4 \pm \sqrt{16 - 12}}{2} = -2 \pm 1 = -1, -3$$

Because the roots are equal, we will have overdamped response.

The conclusion is therefore, as the stiffness decreases, the response will change from underdamped response to critically damped response to overdamped response.

[1 mark for stating the conclusion, 3 marks for calculations and reasoning].

6. Frame $\{B\}$ is initially coincident with frame $\{A\}$. First, $\{B\}$ is rotated about z_A by β degrees, and is subsequently rotated about y_A by α degrees, followed by a rotation about z_A by γ degrees. If the resultant rotation matrix has the following numerical values:

$${}^A_B R = \begin{bmatrix} -0.3062 & -0.9186 & 0.25 \\ 0.8839 & -0.1768 & 0.4330 \\ -0.3536 & 0.3536 & 0.8660 \end{bmatrix}$$

calculate the angles α , β and γ .

[5 marks]

The rotation is represented using the fixed angles representation, as the rotations occur about the original (fixed) axis. In fixed angle representation, the order of matrix multiplication is the opposite of the order of physical rotations:

Physical rotation: Z(beta) then Y(alpha) then Z(gamma).

Rotation matrix:

$$\begin{aligned} {}^A_B R &= R_Z(\gamma) \cdot R_Y(\alpha) \cdot R_Z(\beta) \\ &= \begin{bmatrix} c\gamma & -s\gamma & 0 \\ s\gamma & c\gamma & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} c\alpha & 0 & s\alpha \\ 0 & 1 & 0 \\ -s\alpha & 0 & c\alpha \end{bmatrix} \cdot \begin{bmatrix} c\beta & -s\beta & 0 \\ s\beta & c\beta & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} c\gamma c\alpha & -s\gamma & c\gamma s\alpha \\ s\gamma c\alpha & c\gamma & s\gamma s\alpha \\ -s\alpha & 0 & c\alpha \end{bmatrix} \cdot \begin{bmatrix} c\beta & -s\beta & 0 \\ s\beta & c\beta & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} c\gamma c\alpha c\beta - s\gamma s\beta & -c\gamma c\alpha s\beta - s\gamma c\beta & c\gamma s\alpha \\ s\gamma c\alpha c\beta + c\gamma s\beta & -s\gamma c\alpha s\beta + c\gamma c\beta & s\gamma s\alpha \\ -s\alpha c\beta & s\alpha s\beta & c\alpha \end{bmatrix} \end{aligned}$$

To recover the angles, we note that:

$$c\alpha = 0.8660$$

Thus

$$s\alpha = \pm\sqrt{1 - c^2\alpha} = \pm\sqrt{1 - 0.866^2} = \pm\sqrt{1 - 0.75} = \pm\sqrt{0.25} = \pm 0.5$$

Therefore:

$$\alpha = \text{atan2}(\pm 0.5, 0.8660) = 30^\circ, -30^\circ$$

Next,

$$\begin{aligned} c\gamma s\alpha &= 0.25 \rightarrow c\gamma = \frac{0.25}{s\alpha} \\ s\gamma s\alpha &= 0.4330 \rightarrow s\gamma = \frac{0.4330}{s\alpha} \end{aligned}$$

Therefore:

$$\gamma = \text{atan2}\left(\frac{0.4330}{s\alpha}, \frac{0.25}{s\alpha}\right)$$

For $\alpha = 30^\circ$:

$$\gamma = \text{atan2}\left(\frac{0.4330}{0.5}, \frac{0.25}{0.5}\right) = \text{atan2}(0.8660, 0.5) = 60^\circ$$

For $\alpha = -30^\circ$:

$$\gamma = \text{atan2}\left(\frac{0.4330}{-0.5}, \frac{0.25}{-0.5}\right) = \text{atan2}(-0.8660, -0.5) = 240^\circ$$

Finally:

$$\begin{aligned} -s\alpha c\beta &= -0.3536 \rightarrow c\beta = \frac{0.3536}{s\alpha} \\ s\alpha s\beta &= 0.3536 \rightarrow s\beta = \frac{0.3536}{s\alpha} \end{aligned}$$

Therefore:

$$\beta = \text{atan2}\left(\frac{0.3536}{s\alpha}, \frac{0.3536}{s\alpha}\right)$$

For $\alpha = 30^\circ$:

$$\beta = \text{atan2}\left(\frac{0.3536}{0.5}, \frac{0.3536}{0.5}\right) = \text{atan2}(0.707, 0.707) = 45^\circ$$

For $\alpha = -30^\circ$:

$$\beta = \text{atan2}\left(\frac{0.3536}{-0.5}, \frac{0.3536}{-0.5}\right) = \text{atan2}(-0.707, -0.707) = 225^\circ$$

So the angles are either:

$$\alpha = 30^\circ, \beta = 45^\circ, \gamma = 60^\circ$$

Or:

$$\alpha = -30^\circ, \beta = 225^\circ, \gamma = 240^\circ$$

[3 marks for getting the first quadrant solution, 2 marks for getting the second set of solution]